

Research article

Mapping climate-conditioned PM_{2.5} response under climate change in Taiwan using a high-resolution Geo-AI model

Pei-Yi Wong^{a,b}, Wei-Lun Hsu^c, Chieh-Ying Chen^c, Chien-Fei Chen^d, Chia-Wei Hsu^c,
Shih-Chun Candice Lung^{e,f,g}, Ta-Chih Hsiao^{e,g}, Chia-Pin Yu^h, Wan-Yu Liu^{i,j},
Pau-Chung Chen^{k,l,m,n}, Wei Jie Seow^{o,p}, Chien-Hao Sung^c,
Chih-Da Wu^{c,j,k,q,r,*}

^a Department of Environmental Science and Engineering, National Pingtung University of Science and Technology, Pingtung, Taiwan

^b Center for Sustainable Energy and Climate Change Research and Demonstration, National Pingtung University of Science and Technology, Pingtung, Taiwan

^c Department of Geomatics, National Cheng Kung University, Tainan, Taiwan

^d Department of Sociology, Anthropology and Criminal Justice, Clemson University, South Carolina, United States

^e Research Center for Environmental Changes, Academia Sinica, Taipei, Taiwan

^f Department of Atmospheric Sciences, National Taiwan University, Taipei, Taiwan

^g Graduate Institute of Environmental Engineering, National Taiwan University, Taipei, Taiwan

^h School of Forestry and Resource Conservation, National Taiwan University, Taipei, Taiwan

ⁱ Department of Forestry, National Chung Hsing University, Taichung, Taiwan

^j Innovation and Development Center of Sustainable Agriculture, National Chung Hsing University, Taichung, Taiwan

^k National Institute of Environmental Health Sciences, National Health Research Institutes, Miaoli, Taiwan

^l Institute of Environmental and Occupational Health Sciences, College of Public Health, National Taiwan University, Taipei, Taiwan

^m Department of Public Health, National Taiwan University College of Public Health, Taipei, Taiwan

ⁿ Department of Environmental and Occupational Medicine, National Taiwan University Hospital and National Taiwan University College of Medicine, Taipei, Taiwan

^o Saw Swee Hock School of Public Health, National University of Singapore and National University Health System, Singapore, Singapore

^p Department of Medicine, Yong Loo Lin School of Medicine, National University of Singapore and National University Health System, Singapore, Singapore

^q Research Center for Precision Environmental Medicine, Kaohsiung Medical University, Kaohsiung, Taiwan

^r Chronic Diseases and Health Promotion Research Center, Chang Gung University of Science and Technology, Puji City, Taiwan

ARTICLE INFO

Keywords:

Geospatial artificial intelligence (geo-AI)
Climate-conditioned PM_{2.5} projection
CMIP6 downscaled climate data
IPCC AR6 climate scenarios
Humid subtropical and monsoon-influenced regions
High-resolution air quality modeling

ABSTRACT

This study develops a high-resolution Geo-AI framework to quantify the impact of future climate change on PM_{2.5} concentrations using Taiwan as a subtropical, monsoon-influenced island case. The model integrates long-term ground-based monitoring data (1994–2019), multi-scale geo-environmental predictors, and statistically down-scaled CMIP6 meteorology, implemented using a Gradient Boosting Machine. The resulting model demonstrates strong predictive performance ($R^2 = 0.81$ and $RMSE = 8.69 \mu\text{g}/\text{m}^3$) and effectively captures PM_{2.5} dynamics within complex islands and coastal environments. By explicitly coupling a Geo-AI model with Intergovernmental Panel on Climate Change (IPCC) Sixth Assessment Report (AR6) climate scenarios, this study extends data-driven PM_{2.5} modeling from historical estimation to climate-conditioned future projection, addressing a key methodological gap in existing air-quality research. SHAP-based interpretability analysis identifies temperature and precipitation as dominant predictors, underscoring their central role in shaping future aerosol variability. The SHAP results further indicate that both temperature and precipitation exhibit nonlinear relationships across different temporal and regional scales and overall inverse associations with PM_{2.5} concentrations, clarifying the climate-driven effects of warming and hydrological change on PM_{2.5} dynamics under humid subtropical conditions. Across four Shared Socioeconomic Pathway scenarios, projected PM_{2.5} concentrations consistently decline in the near and midterm (between -1.25 and $-1.5 \mu\text{g}/\text{m}^3$), followed by increasing spatial heterogeneity in the long term, with localized PM_{2.5} hotspots emerging under severe warming conditions. These findings suggest that climate change may generate uneven air-quality responses across space, highlighting the limitations of regional mean assessments and the need for high-resolution, climate-informed mitigation and adaptation

* Corresponding author. Department of Geomatics, National Cheng Kung University, No.1, University Road, Tainan City, 701, Taiwan.

E-mail address: chidawu@mail.ncku.edu.tw (C.-D. Wu).

strategies. The proposed framework provides a transferable tool for climate-responsive air-quality planning in humid subtropical, monsoon-influenced, and densely populated regions worldwide.

1. Introduction

Climate change and extreme weather events are increasingly recognized as major drivers of future fine particulate matter (PM_{2.5}) concentrations through their influence on meteorological pathways. Changes in temperature, precipitation, boundary-layer height, humidity, wind fields, and atmospheric stability can fundamentally affect the formation, transport, accumulation, and removal of airborne particles (Megaritis et al., 2014; Yang et al., 2016). However, the direction and magnitude of meteorological impacts on PM_{2.5} vary substantially across regions. For example, in some temperate regions, warming may enhance photochemical reactions and secondary aerosol formation, leading to higher PM_{2.5} concentrations. In contrast, in tropical or subtropical regions such as Taiwan, elevated temperatures are often associated with enhanced boundary-layer mixing or altered precipitation regimes, which can reduce PM_{2.5} accumulation (Feng et al., 2019; Liu et al., 2021; Kumar et al., 2024a, 2024b, 2025a, 2025b, Kumar and Choudhary, 2025). These contrasting responses underscore the need for region-specific assessments that explicitly account for local climate conditions, geographical context, and emission structures, rather than assuming uniform global effects.

Although the adverse health impacts of PM_{2.5} exposure are well established, robust estimation of future health burdens requires reliable projections of PM_{2.5} concentrations under climate change that capture spatial heterogeneity, seasonal variability, and the potential for extreme pollution events. Most existing studies addressing future PM_{2.5} have primarily emphasized emission scenarios or relied on chemical transport models (CTMs), which explicitly represents atmospheric dynamics, emissions, and chemical reactions, allowing for causal interpretation and scenario-based analysis under different emission and climate conditions (West et al., 2013). While CTMs provide physically based simulations, they depend on detailed emission inventories, complex parameterizations, and substantial computational resources, which constrain their applicability for high-resolution, multi-scenario projections, particularly over regions with complex terrain (Lyu et al., 2025).

Geospatial artificial intelligence (Geo-AI) has emerged as a promising approach for air-quality modeling because it can integrate long-term monitoring records, land-use characteristics, and meteorological variables to capture highly nonlinear relationships between pollution and climate with a high spatiotemporal resolution (Amnuaylojaroen et al., 2022). However, Geo-AI is generally limited in interpretability, as the learned relationships are statistical rather than mechanistic, making it difficult to directly infer physical causality. To date, most Geo-AI applications have emphasized historical reconstruction or near real-time mapping of PM_{2.5}, with the primary goal of improving retrospective estimation accuracy (Lee et al., 2019; Wong et al., 2022). Extending Geo-AI to future air-quality simulation remains limited, largely because data-driven models are seldom coupled with long-term climate projections and explicit climate-change scenarios. As a result, the potential of Geo-AI for future climate-conditioned PM_{2.5} projection has not yet been fully realized, and Geo-AI has not been widely adopted as a forward-looking tool for climate change-informed air quality assessment.

Taiwan is a subtropical island in East Asia characterized by complex topography, strong monsoonal circulation, and distinct spatial heterogeneity. The Central Mountain Range significantly influences airflow, precipitation, and pollutant dispersion, resulting in clear contrasts between the highly urbanized and industrialized western plains and the sparsely populated mountainous eastern region. Over recent decades, rising temperatures and increasing climate variability, including more

frequent extreme rainfall and drought events, have further altered atmospheric processes. These characteristics make Taiwan an ideal natural laboratory for investigating PM_{2.5} dynamics and assessing climate-driven air quality risks. Insights from Taiwan therefore have relevance beyond the local context, offering lessons for other densely populated, monsoon-influenced regions across Asia that face similar climatic and geographic constraints.

Previous studies have suggested that socioeconomic changes, such as population growth, industrial restructuring, and land-use evolution, may indirectly influence climate and meteorological projections (Feng et al., 2019). To the extent that these influences are already embedded within modeled temperature and precipitation trajectories, explicitly incorporating future socioeconomic variables may be redundant and may introduce additional uncertainty. Moreover, given the limited availability and high uncertainty of future emissions inventories, this study prioritizes meteorological mechanisms linking climate change to PM_{2.5} dynamics rather than explicitly modeling future emissions.

To address these gaps, we develop a high-resolution Geo-AI framework that integrates long-term ground-based PM_{2.5} observations, multi-scale geo-environmental predictors, and statistically downscaled climate projections consistent with the Intergovernmental Panel on Climate Change (IPCC) Sixth Assessment Report (AR6). Moving beyond Geo-AI studies focused on historical estimation, this framework advances data-driven air-quality modeling toward climate-conditioned projection under multiple Shared Socioeconomic Pathway (SSP) scenarios. Specifically, this study aims to: (1) integrate a high-resolution Geo-AI model with IPCC AR6 climate scenarios to simulate future PM_{2.5} concentration changes, therefore extending Geo-AI applications from retrospective mapping to forward-looking, climate-conditioned air-quality projection; and (2) disentangle the climate-driven effects of temperature and precipitation on future PM_{2.5} dynamics using SHAP-based interpretability analysis, clarifying how warming and hydrological changes may modulate PM_{2.5} concentrations under climate change. Using Taiwan as a subtropical, monsoon-influenced island case, the proposed framework provides high-resolution insights into how climate change may reshape PM_{2.5} spatial patterns and associated air-quality risks, and it offers a transferable methodological blueprint for climate-responsive air-quality assessment in humid subtropical, monsoon-influenced, and densely populated regions worldwide.

2. Materials and methods

2.1. Baseline air quality and geo-environmental database

This study compiled a comprehensive baseline database consisting of historical PM_{2.5} concentrations and a diverse set of geo-environmental predictors to identify key drivers of air pollution and to train the Geo-AI model. These predictors were subsequently integrated with future climate projections to estimate PM_{2.5} concentrations under climate-change scenarios.

Daily PM_{2.5} concentrations from 1994 to 2019 were obtained from 74 air-quality monitoring stations operated by the Ministry of Environment (MOE) (Fig. 1). Because nationwide PM_{2.5} monitoring was not established until 2006, PM_{2.5} levels for 1994–2005 were reconstructed from PM₁₀ measurements using a station-specific linear regression approach. This method has been widely applied for historical PM_{2.5} back-estimation in East Asia and has demonstrated robust estimation in prior studies (Wong et al., 2023).

The predictors encompassed meteorological parameters, land use and land cover (LULC), special emission sources, landmarks, satellite-derived vegetation indices, digital terrain model (DTM) variables,

population density, and seasonal factors (Table A.1). Daily meteorological data including temperature, precipitation, humidity, and wind fields were obtained from the Taiwan Central Weather Administration. LULC predictors included spatial densities of forest, farmland, orchards, water bodies, residential areas, and industrial or manufacturing sites, which are commonly associated with pollution emissions (Li et al., 2023; Wong et al., 2021). The LULC data was obtained from the National Land Surveying and Mapping Center, Ministry of the Interior with survey periods of 2006, 2015, and 2017.

To better represent local anthropogenic emissions, this study incorporated detailed point-of-interest (POI) information including temples, Taiwanese restaurants, waste-incineration facilities, and thermal power plants for the year 2008 and 2010. POI-based proxies have been demonstrated to capture localized combustion activities and non-industrial emission sources that are often absent from conventional emission inventories (Hsu et al., 2023; Wong et al., 2022). Geomorphological predictors including elevation, slope, and aspect were extracted from the DTM to characterize terrain-induced airflow and pollutant dispersion patterns, which are particularly influential in mountainous environments such as Taiwan (Chen et al., 2023; Lai and

Lin, 2020). Vegetation effects were represented using the Normalized Difference Vegetation Index (NDVI) derived from the Moderate Resolution Imaging Spectroradiometer (MODIS) provided by National Aeronautics and Space Administration (NASA). NDVI has been widely recognized for its role in $PM_{2.5}$ removal via dry deposition and air-vegetation interactions (Li et al., 2023; Srbinovska et al., 2021). Population density was also incorporated as a proxy for anthropogenic activity intensity and associated emissions, a commonly adopted approach in high-density urban regions (Babaan et al., 2023; Wong et al., 2024a).

All predictors were spatially processed using multi-scale circular buffers surrounding each monitoring station, with radii of 50 m to 5000m (50, 150, 250, 500, 750, 1,000, 1,250, 1,500, 1,750, 2,000, 2,500, 3,000, 4,000, and 5,000m) following established Geo-AI modeling practices (Babaan et al., 2023; Hsu et al., 2023; Wong et al., 2024a). For predictors characterized by relatively few sources but potentially large impacts, such as airports, thermal power plants, and incineration facilities—the nearest-distance metric between each source and monitoring station was additionally calculated. This approach follows best practices for representing low-frequency, high-impact point

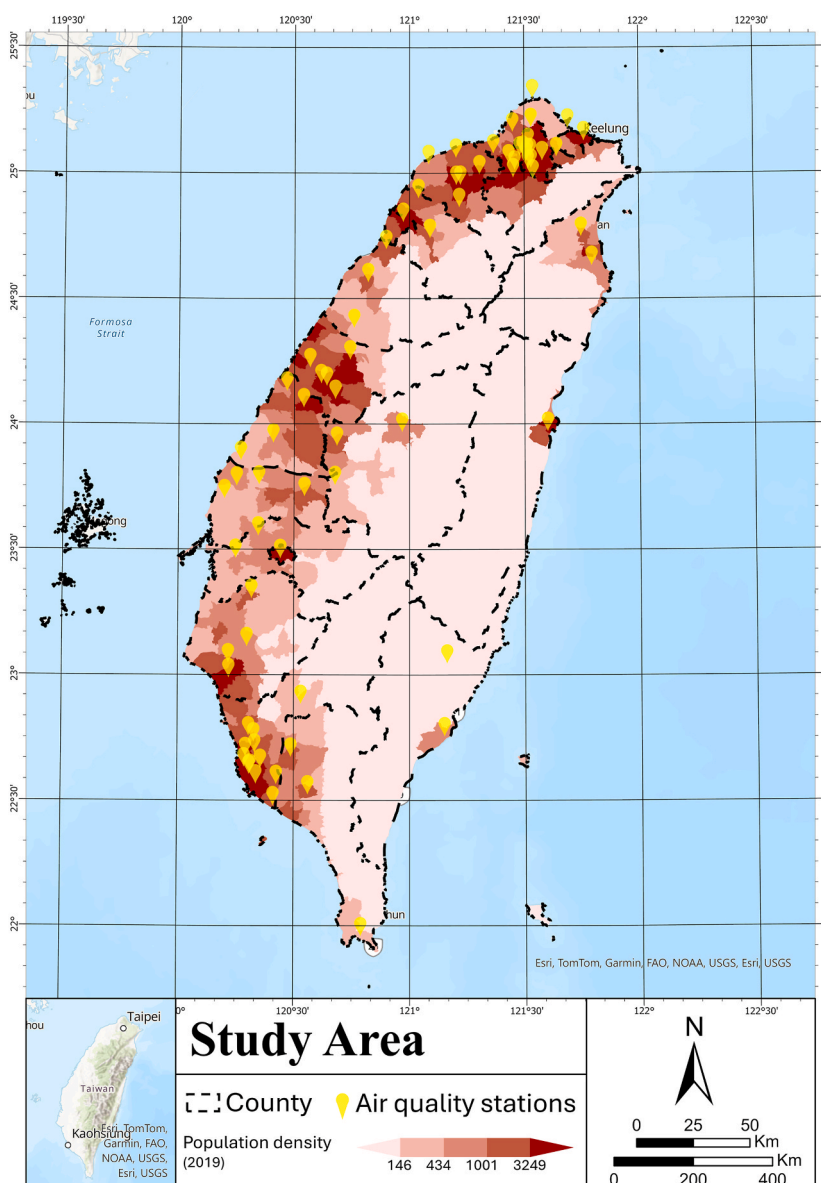


Fig. 1. Distribution of air quality monitoring stations and township population density.

emitters in air-quality models (Amnuaylojaroen et al., 2022; Wong et al., 2022). Together, this integrated spatial-temporal database provided a robust foundation for training the Geo-AI model and for evaluating how climate-driven changes in meteorological conditions may translate into future PM_{2.5} variations across Taiwan.

2.2. Future temperature and precipitation database

Future projections of temperature and precipitation under multiple greenhouse gas emission scenarios were obtained from the Taiwan Climate Change Projection Information and Adaptation Knowledge Platform (TCCIP). These projections were based on global climate models (GCMs), which simulate long-term atmospheric dynamics and radiative forcing mechanisms under future socioeconomic pathways. Although GCMs provide comprehensive Earth system simulations, their coarse native spatial resolution limits application in small and topographically complex regions such as Taiwan (Döscher et al., 2021; Demuzere et al., 2022).

To overcome this limitation, TCCIP applied a statistical downscaling framework to refine GCM outputs to a spatial resolution of $0.05^\circ \times 0.05^\circ$ (approximately 5 km) (Lin et al., 2022). The downscaled datasets were generated using outputs from 31 GCMs developed by 21 international climate research institutions, covering both a historical baseline period (1960–2014) and a future projection period (2015–2100) at daily temporal resolution (refer to the supplementary materials). Previous studies have demonstrated that statistical downscaling substantially improves the representation of local climate conditions and terrain-induced variability, particularly in mountainous island environments in East Asia (Lovato et al., 2022; Peng et al., 2019).

This study selected three GCMs, TaiESM1, MIROC6, and MPI-ESM1-2-LR, based on their documented performance in simulating East Asian monsoon systems and regional hydroclimate characteristics (Peng et al., 2019; Wang et al., 2021a,b). TCCIP provides projections across four Shared Socioeconomic Pathway (SSP) scenarios: SSP1–2.6, SSP2–4.5, SSP3–7.0, and SSP5–8.5, consistent with the IPCC AR6. These scenarios

span a wide range of future socioeconomic development and emissions trajectories, from low-emission sustainable development (SSP1–2.6) to high emission and fossil-fuel intensive trajectories (SSP5–8.5) (O'Neill et al., 2016).

For analytical purposes, future climate projections were grouped into three-time horizons: 2021–2040 (near-term), 2041–2060 (mid-term), and 2081–2100 (long-term). The use of multiple climate models, emissions scenarios, and temporal windows allows for a robust assessment of how long-term changes in temperature and precipitation may influence PM_{2.5} concentrations under different warming pathways. Because aerosol formation, transport, removal, and atmospheric stagnation are strongly modulated by regional hydroclimatic conditions (Ma et al., 2022; Westervelt et al., 2016), this approach enables the quantification of climate-driven impacts of PM_{2.5} changes without reliance on highly uncertain future emissions inventories.

2.3. Development and validation of the Geo-AI modeling framework

This study developed a Geo-AI modeling framework (Fig. 2) to estimate PM_{2.5} concentrations using five widely adopted machine-learning algorithms: eXtreme Gradient Boosting (XGBoost), Gradient Boosting Machine (GBM), Light Gradient Boosting Machine (LightGBM), Categorical Boosting (CatBoost), and Random Forest (RF). These algorithms have been extensively applied in air-quality and particulate matter estimation due to their ability to capture non-linear relationships, accommodate heterogeneous predictor types, and model complex feature interactions (Babaan et al., 2023; Hsu et al., 2023; Wong et al., 2024a, 2024b). The detailed hyperparameter configurations for each algorithm are provided in Table A2. The hyperparameter tuning procedure was conducted using AutoML (package: auto_ml) with a grid search method in python.

Model inputs consisted of influential predictors identified through SHAP-based feature selection, as described in Section 3.4. Model performance was evaluated using five complementary statistical metrics. Predictive accuracy was quantified using the coefficient of

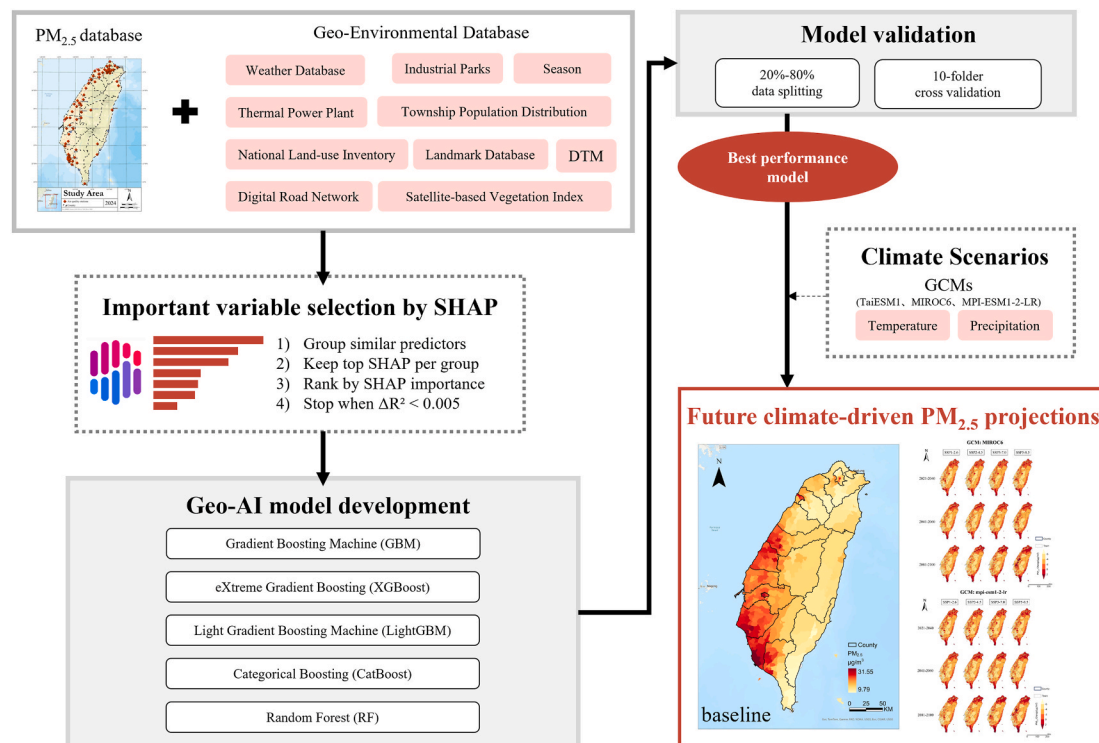


Fig. 2. Workflow of the Geo-AI modeling framework for PM_{2.5} estimation, predictor selection, model validation, and climate-conditioned future projection under CMIP6 SSP scenarios.

determination (R^2) and adjusted R^2 , while prediction errors were assessed using Root Mean Square Error (RMSE), Mean Square Error (MSE), and Mean Absolute Error (MAE). The full dataset was randomly partitioned into training (80%) and independent testing (20%) subsets. In addition, a 10-fold cross-validation (CV) procedure was implemented, in which 90% of the data were used for model training and the remaining 10% for validation in each iteration. This process was repeated ten times to ensure that all data subsets were systematically evaluated.

To identify the optimal Geo-AI model while minimizing overfitting, a multi-stage model selection strategy was applied. As an initial screening criterion, only models achieving a training R^2 greater than 0.80 were retained for further evaluation. Model robustness was then assessed using two complementary indicators: (1) the discrepancy between training and testing R^2 , and (2) the discrepancy between training and 10-fold CV R^2 . These indicators were used to evaluate model generalization ability and sensitivity to data partitioning. The algorithm demonstrating the most stable and balanced performance across these criteria was ultimately selected as the final Geo-AI model for $PM_{2.5}$ prediction.

To further assess model robustness and generalizability across varying spatiotemporal conditions, both temporal and spatial validation analyses were conducted. Temporal validation included year-by-year evaluations from 1994 to 2019, as well as seasonal assessments for spring, summer, autumn, and winter. Spatial validation was performed across seven officially defined air-quality management zones in Taiwan—Northern, Hsinchu-Miaoli, Central, Yunlin-Chiayi-Tainan, Kaohsiung-Pingtung, Yilan, and Hualien-Taitung—thereby testing model applicability across regions characterized by distinct emission profiles, climatic conditions, and terrain complexity.

2.4. Selection of key predictors

Following the construction of the baseline database, this study used SHapley Additive exPlanations (SHAP) to quantify the marginal contribution of each predictor to $PM_{2.5}$ estimation. SHAP is a game-theoretic interpretability framework that decomposes ML predictions into additive feature-level contributions, allowing for consistent and comparable assessment of predictor importance across complex, nonlinear models (Lundberg and Lee, 2017). This approach is particularly advantageous in climate-air quality research, where meteorological influences on particulate matter often exhibit nonlinear responses, interactions, and threshold effects that are poorly captured by traditional linear approaches.

Initial SHAP values were derived from preliminary ML models that included the full set of candidate variables. To reduce redundancy and mitigate multicollinearity, variables with similar conceptual or spatial characteristics (e.g., NDVI calculated using different buffer distances) were grouped, and only the variable with the highest SHAP importance within each group was retained. The remaining predictors were then ranked according to their SHAP contributions and sequentially introduced into the modeling framework.

An iterative, stepwise addition procedure was used to determine the optimal predictor set. Model performance was evaluated at each step, and variable inclusion was terminated when the moving-average improvement in explanatory power (R^2) fell below 0.005 for three consecutive additions, indicating diminishing marginal gains (Hsu et al., 2023; Wong et al., 2024a). This data-driven criterion ensured that only predictors with substantive explanatory value to $PM_{2.5}$ were retained, while avoiding unnecessary model complexity and overfitting.

Notably, climatic variables including temperature and precipitation emerged as high-ranked predictors based solely on their SHAP contributions, underscoring their substantial influence on $PM_{2.5}$ variability. Their inclusion was therefore not imposed a priori but arose directly from empirical evidence, reinforcing the central role of meteorological forcing in shaping particulate pollution dynamics under future climate

scenarios. In this context, SHAP functioned as not only a variable-selection and dimensionality reduction tool, but also as a mechanistic inference framework for identifying climate-sensitive drivers of $PM_{2.5}$ concentrations.

2.5. Future $PM_{2.5}$ projections and climate-driven spatial changes

The finalized Geo-AI model was applied to project future $PM_{2.5}$ concentrations using projected temperature and precipitation projections from three AR6-compatible GCMs: TaiESM1, MIROC6, and MPI-ESM1-2-LR. These models were selected based on their demonstrated ability in simulating climate dynamics over the Western North Pacific and East Asian Monsoon region, within which Taiwan is located. TaiESM1, in particular, was developed to represent Taiwan and surrounding East Asia, and offers improved characterization of complex island topography, land-sea thermal contrasts, and orographic rainfall processes (Peng et al., 2019; Wang et al., 2021a,b). Meanwhile, MIROC6 and MPI-ESM1-2-LR have similarly demonstrated strong performance in capturing large-scale monsoon circulation, precipitation variability, and hydro-climatic seasonality in East Asia, supporting their applicability for future air-quality assessments in the region (Chen and Jin, 2021). Using multiple GCMs further enables partial representation of structural and parametric uncertainty in climate predictions, therefore enhancing the robustness of the projection projected $PM_{2.5}$ outcomes (TCCIP, 2024).

$PM_{2.5}$ predictions were estimated for a historical baseline period (1994–2019) and three future time period: near-term (2021–2040), mid-term (2041–2060), and long-term (2081–2100) conditions under four SSP scenarios (SSP1-2.6, SSP2-4.5, SSP3-7.0, and SSP5-8.5). Baseline $PM_{2.5}$ concentration maps were derived using the final set of influential predictors identified from model training. Future $PM_{2.5}$ concentrations were simulated by replacing historical meteorological inputs with downscaled temperature and precipitation projections, while all geo-environmental predictors (e.g., land use, elevation, emission sources, vegetation cover) were kept constant to isolate the effects of climate-change forcing.

To assess climate-driven changes in future air quality, difference maps were created by subtracting baseline $PM_{2.5}$ estimates for each GCM-SSP-time-horizon combination. These spatial comparisons enabled quantitative assessment of how $PM_{2.5}$ concentrations may increase or decrease across Taiwan under alternative warming pathways. The resulting differential patterns were used to identify (i) regions experiencing net $PM_{2.5}$ reductions, indicating potential climate-related air-quality benefits, and (ii) localized hotspots where $PM_{2.5}$ concentrations may increase despite broader declines. Such areas represent locations where climate adaptation strategies and targeted emission-control policies may be necessary to mitigate future air-quality risks.

3. Results

3.1. Baseline trends of $PM_{2.5}$, temperature, and precipitation (1994–2019)

During the baseline period (1994–2019), the annual mean $PM_{2.5}$ concentration across Taiwan was $31.23 \mu\text{g}/\text{m}^3$ with value ranging from 7.64 to $81.96 \mu\text{g}/\text{m}^3$ (Table 1). A pronounced downward trend in $PM_{2.5}$ was observed over time (p -value < 0.001 through Mann-Kendall test) (Fig. 3a), with annual averages declining from $38.94 \mu\text{g}/\text{m}^3$ in the mid-1990s to $19.71 \mu\text{g}/\text{m}^3$ by 2019. Over the same period, the annual mean near-surface temperature averaged 22.79°C , with observed values ranging from 17.00 to 25.86°C . A consistent warming trend was evident (p -value < 0.001) (Fig. 3b), with the warmest year exhibiting a mean temperature approximately 2.16°C higher than that of the coolest year during the study period.

Annual precipitation exhibited substantial interannual variability, with a long-term mean of 2207.13 mm and a wide range spanning from 595.60 to 3426.00 mm (Fig. 3c). In contrast to $PM_{2.5}$ and temperature,

Table 1

Descriptive statistics of annual mean PM_{2.5} concentration, near-surface temperature, and precipitation in Taiwan during the baseline period (1994–2019).

Variable	Unit	Mean ± SD	Min	Max	Q1	Median	Q3
PM _{2.5}	µg/m ³	31.23 ± 7.50	7.64	81.96	23.91	30.38	38.11
Temperature	°C	22.79 ± 1.48	17.00	25.86	21.90	22.92	23.82
Precipitation	mm	2207.13 ± 833.41	595.60	3426.00	1599.69	2106.92	2630.81

Note: Values represent minimum, maximum, mean ± standard deviation (SD), and quartiles (Q1, median, Q3).

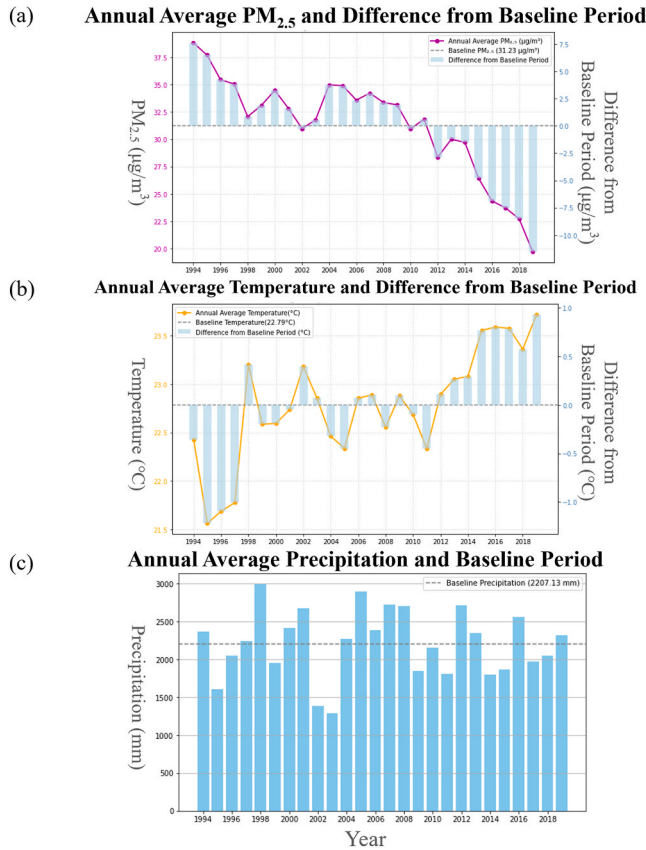


Fig. 3. Annual trends of (a) PM_{2.5} concentration, (b) near-surface temperature, and (c) precipitation in Taiwan from 1994 to 2019. Mean values for each year are shown, illustrating long-term variations across the baseline period.

precipitation did not exhibit a clear monotonic increasing or decreasing trend over the baseline period. Collectively, these baseline observations indicate a sustained decline in PM_{2.5} concentrations alongside a steady rise in near-surface temperature and highly variable precipitation conditions between 1994 and 2019. This historical pattern provides a critical reference framework for evaluating future climate-driven changes in air quality under different warming scenarios.

3.2. Performance evaluation and validation of the Geo-AI model

The training R^2 for XGBoost, GBM, LGBM, CatBoost, and Random Forest, were 0.93, 0.81, 0.64, 0.58, and 0.78, respectively (Fig. 4). The corresponding testing R^2 were 0.61, 0.60, 0.60, 0.57, and 0.59; and the 10-fold CV R^2 were 0.60, 0.60, 0.59, 0.57, and 0.59, respectively. Models with a training R^2 lower than 0.80 were excluded, leaving only XGBoost and GBM for further evaluation. For XGBoost, the training–testing R^2 gap was 0.32, and the training–10-fold CV R^2 gap was 0.33. In contrast, GBM exhibited smaller discrepancies of 0.21 for both indicators, suggesting a lower risk of overfitting and more stable generalization performance. Based on this comparative assessment, the GBM algorithm was selected as the final Geo-AI model and applied in

subsequent PM_{2.5} predictions.

For the selected GBM model, the RMSE, MSE, and MAE of the training model were 8.69, 75.43, and 4.37 µg/m³, respectively. For the independent testing dataset, the corresponding values increased to 12.27, 150.51, and 6.57 µg/m³. Similar error magnitudes were observed under 10-fold CV, with RMSE, MSE, and MAE were 12.32, 151.86, and 6.55 µg/m³, respectively (Fig. 4), further indicating stable predictive performance. To assess model robustness across spatiotemporal domains, both temporal and spatial validation were conducted. Temporal validation included annual and seasonal assessments. The annual validation yielded an average R^2 of 0.80, with values ranging from 0.75 to 0.84 (Figure A1a), while seasonal validation produced an average R^2 of 0.78, ranging from 0.76 to 0.80 (Figure A1b). Spatial validation was performed across official air-quality zones in Taiwan, resulting in an average R^2 of 0.75, with values spanning 0.62 to 0.85 (Figure A1c). The average prediction error (RMSE) was 7.94 µg/m³, ranging from 5.62 to 9.81 µg/m³. Overall, the predicted values are highly consistent with the observed measurements, with errors ranging between +0.5 and −1.5 (Fig. 5). This close agreement and minimal deviation indicate that the model effectively captures the variability of PM_{2.5} concentrations. These results demonstrate that the GBM-based Geo-AI model provides robust and consistent PM_{2.5} predictions across different years, seasons, and regions. Its strong generalization performance supports its suitability for projecting future PM_{2.5} concentrations under climate-driven scenarios.

3.3. Influence of climate and environmental predictors on PM_{2.5} variability

SHAP analysis identified ten influential predictors in the final GBM model, with precipitation and temperature emerging as the two most important climate-related drivers of PM_{2.5} variability. Fig. 6 presents the ranked importance of all selected predictors, which include (in descending order): precipitation, X-coordinate, temperature, population density, summer seasonality, normalized difference vegetation index (NDVI), distance to airports, density of Taiwanese restaurants, spring seasonality, and forest coverage. Among these, precipitation exhibited the highest overall SHAP importance, followed by temperature as the dominant thermal predictor.

The SHAP dependence patterns further reveal the directionality of predictor effects. Precipitation, NDVI, forest coverage, summer, and distance to airports were predominantly associated with negative SHAP values, indicating that higher values of these variables tended to reduce predicted PM_{2.5} concentrations. In contrast, population density showed consistently positive SHAP values, reflecting its role as a proxy for anthropogenic emission intensity. Temperature exhibited high SHAP importance with both positive and negative values, suggesting a nonlinear and context-dependent influence on PM_{2.5} levels, likely mediated by competing processes such as enhanced atmospheric mixing and increased secondary aerosol formation.

Future climate-driven PM_{2.5} projections based on the three GCMs indicate consistent reductions in PM_{2.5} concentrations relative to the baseline across all four SSP scenarios (Fig. 7a). The magnitude of these reductions was greatest in the near-term (2021–2040) and mid-term (2041–2060) periods, but diminished in the long-term (2081–2100), particularly under higher-emission scenarios (SSP3-7.0 and SSP5-8.5). Concurrently, projected near-surface temperatures increased across all SSP scenarios and time horizons (Fig. 7b), while precipitation generally

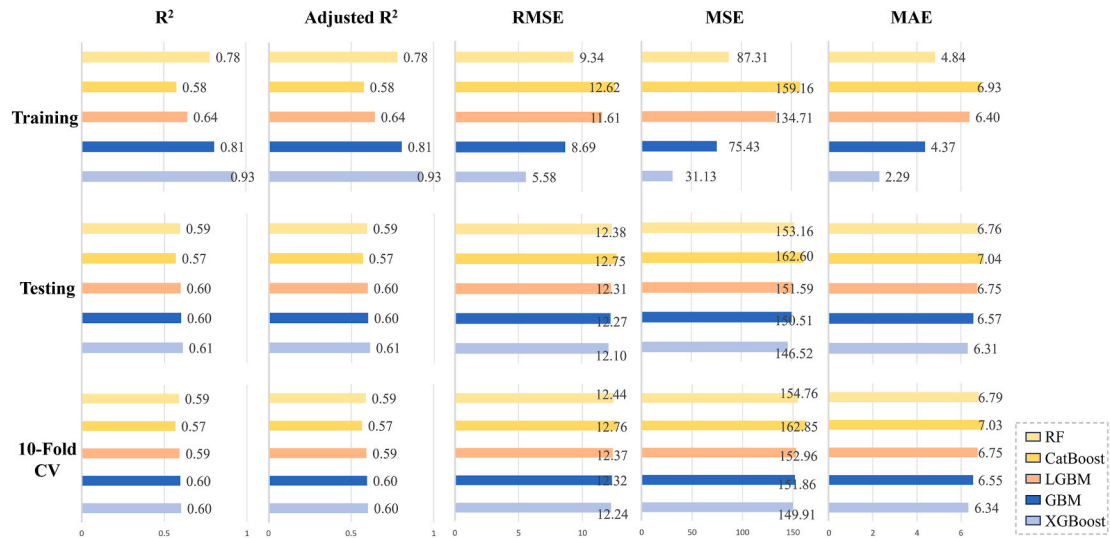


Fig. 4. Performance comparison of five machine-learning algorithms (XGBoost, GBM, LGBM, CatBoost, and Random Forest) based on training, testing, and 10-fold cross-validation (CV) datasets. Metrics include R², RMSE, MSE, and MAE. GBM demonstrated the most balanced performance and was selected as the final Geo-AI model.

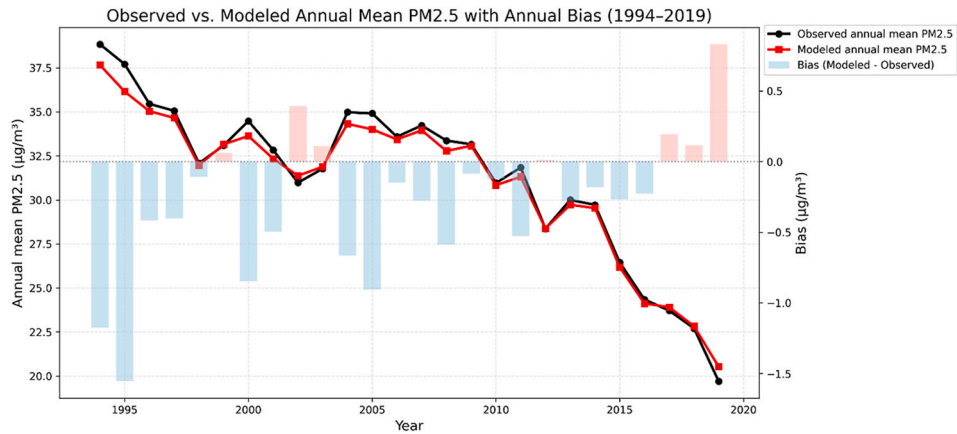


Fig. 5. Time series plot of observed vs. modeled annual mean PM_{2.5} over 1994–2019.

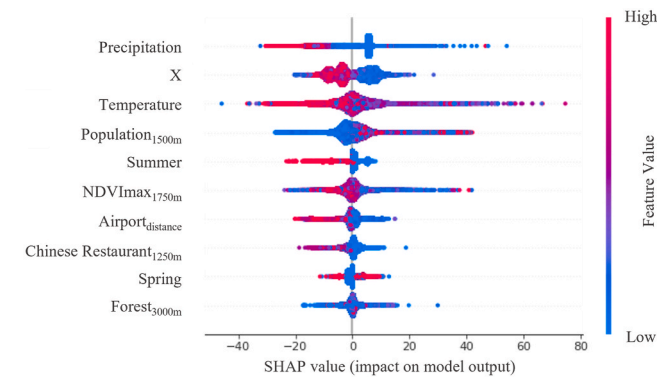


Fig. 6. SHAP summary plot showing the distribution of feature importance for the final GBM model. Variables located higher on the y-axis have stronger influence on PM_{2.5} prediction. Colors represent the standardized values of each predictor (red = high; blue = low), and SHAP values indicate contribution magnitude to predicted PM_{2.5} levels.

declined, with the magnitude of change varying by scenario and period (Fig. 7c). Inter-model comparisons further reveal systematic differences

in projected climate forcing. Among the three GCMs, TaiESM1 consistently projected higher temperatures and lower precipitation relative to MIROC6 and MPI-ESM1-2-LR. These differences contribute to inter-model variability in PM_{2.5} projections and highlight the sensitivity of future air-quality outcomes to regional climate model representations.

3.4. Spatial distribution of future PM_{2.5} concentrations across climate scenarios

Future PM_{2.5} concentrations were spatially estimated for Taiwan under four SSP climate scenarios (SSP1-2.6, SSP2-4.5, SSP3-7.0, and SSP5-8.5) for three projection periods: near-term (2021–2040), mid-term (2041–2060), and long-term (2081–2100) (Figure A2). During the baseline period (1994–2019), the average PM_{2.5} level was 15.18 µg/m³, with higher concentrations observed across southwestern Taiwan and lower levels across eastern mountainous regions (Figure A3). Across all SSP scenarios, future PM_{2.5} concentrations show an overall decrease relative to the baseline period. However, the magnitude and continuity of the decreasing trend vary by scenario. Under SSP1-2.6, PM_{2.5} levels remain lower than the baseline but show slightly higher values in the mid-term compared to the near-term. Under SSP2-4.5, concentrations decrease in the near-term and mid-term and subsequently increase in the long-term. Under SSP3-7.0 and SSP5-8.5, PM_{2.5} shows an increasing

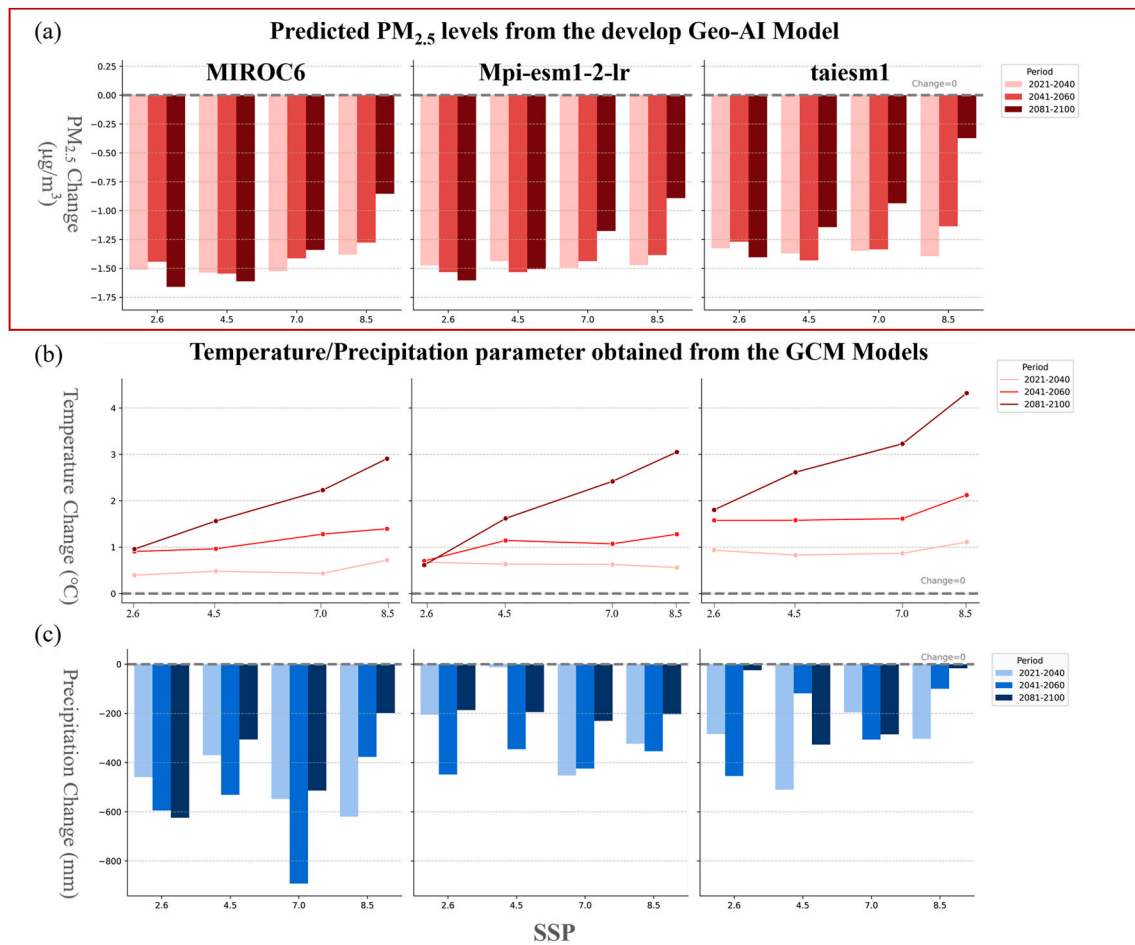


Fig. 7. Relative changes in (a) PM_{2.5} concentration, (b) near-surface temperature, and (c) precipitation under four SSP climate scenarios (SSP1-2.6, SSP2-4.5, SSP3-7.0, SSP5-8.5), compared to the baseline period (1994–2019). Bars represent mean projections across three global climate models (TaiESM1, MIROC6, and MPI-ESM1-2-LR) for near-term (2021–2040), mid-term (2041–2060), and long-term (2081–2100) periods.

trend across all three future periods.

Difference maps further illustrate spatial shifts in PM_{2.5} concentrations between the baseline and future periods (Fig. 7). Across all scenarios, most townships exhibit lower concentrations than the baseline, with larger reductions occurring in western and southwestern regions. In contrast, some increases are detected in northeastern and southern Taiwan. The maximum projected reduction is approximately 6 μg/m³, whereas localized increases are projected up to approximately 4 μg/m³. Under more severe scenarios and in the long-term period, the number of townships exhibiting PM_{2.5} levels above the baseline increases. These spatial projections, consistently observed across TaiESM1, MIROC6, and MPI-ESM1-2-LR outputs (Figure A4–A.6), highlight contrasting regional responses of PM_{2.5} to projected climate forcing across Taiwan. To address the potential masking effect of extreme events, we have expanded the analysis to investigate the spatial distribution of 95th percentile PM_{2.5} concentrations under baseline period (Figure A7) and future scenarios (Figure A8 to A.10). Changes in the 95th percentile PM_{2.5} concentrations and comparison between historical and future periods were also discussed in Figure A11 to A.13. The baseline map shows that the 95th PM_{2.5} specifically concentrated in southern Taiwan. Using different GCMs to generate the 95th percentile PM_{2.5} concentration maps reveal that elevated concentrations are also primarily concentrated in southern regions. As the severity of SSP scenarios increases, the spatial distribution of high PM_{2.5} concentrations gradually shifts toward central regions. In terms of the differences between future and baseline PM_{2.5} concentrations, the results indicate that areas with originally high concentrations in southern regions exhibit more

pronounced reductions, while only a limited number of townships across Taiwan show increases in future PM_{2.5} levels.

4. Discussion

4.1. Summary of key findings

This study developed a high-resolution Geo-AI framework that integrates long-term ground-based observations, multi-scale geo-environmental predictors, and statistically downscaled CMIP6 climate projections to assess future PM_{2.5} concentrations in Taiwan under IPCC AR6 Shared Socioeconomic Pathway (SSP) scenarios. A central contribution of this work lies not only in its technical integration of Geo-AI and climate projections, but in demonstrating that data-driven air-quality models can be meaningfully extended from retrospective mapping to climate-conditioned forward projection. By explicitly isolating the meteorological pathway—particularly temperature and precipitation—this framework addresses a key methodological gap in the current Geo-AI literature, which has largely focused on historical estimation rather than future climate responsiveness.

Regarding model performance, the training R² was 0.80, but the test R² was 0.60, which indicated moderate predictive performance. Some machine learning-based studies on PM_{2.5} estimation have reported model performance generally in the range of R² ≈ 0.5–0.7 (Donkelaar et al., 2014; Xiao et al., 2020; Zaman et al., 2021), especially in Malaysia, where the climate condition is similar to Taiwan. These studies indicate that the testing model performance reported in this

study ($R^2 \approx 0.60$) is consistent with previously published work, particularly for high-resolution, large-area $PM_{2.5}$ modeling under heterogeneous environmental conditions. Given the complexity of $PM_{2.5}$ processes and long-term prediction, this level of performance is considered acceptable.

Model results consistently highlight precipitation and temperature as dominant predictors of $PM_{2.5}$ variability, alongside population density and vegetation-related factors. This finding aligns with previous studies emphasizing the importance of meteorological controls on particulate pollution under climate change, while further demonstrating that climate-driven forcing alone can substantially reshape $PM_{2.5}$ spatial patterns even under constant-emission assumptions. The SHAP-based interpretability analysis reveals predominantly inverse associations between $PM_{2.5}$ and both precipitation and temperature, reflecting the combined influence of wet scavenging, enhanced atmospheric mixing, and climate-modulated dispersion processes in humid subtropical environments.

Temperature influences $PM_{2.5}$ concentrations primarily through its effects on atmospheric stability and boundary layer dynamics. Higher temperatures generally enhance atmospheric convection and increase boundary layer height, which facilitates the vertical dispersion and dilution of pollutants, thereby reducing near-surface $PM_{2.5}$ concentrations (Luo et al., 2025). In addition, temperature affects photochemical reaction rates and secondary aerosol formation, leading to complex and nonlinear responses depending on environmental conditions. Precipitation plays a critical role in $PM_{2.5}$ removal through wet deposition (scavenging processes), which efficiently removes particulate matter from the atmosphere. Increased precipitation is therefore typically associated with lower $PM_{2.5}$ concentrations. However, the effectiveness of wet scavenging depends on precipitation intensity and frequency, as well as aerosol properties.

In the context of Taiwan, these processes are further modulated by strong monsoonal circulation and complex topography. During the winter monsoon, stagnant atmospheric conditions and long-range transport from continental sources can enhance $PM_{2.5}$ accumulation. In contrast, summer monsoon conditions, characterized by higher temperature, stronger convection, and increased precipitation, generally promote pollutant dispersion and removal, resulting in lower $PM_{2.5}$ levels (Chuang et al., 2017; Lee et al., 2019).

Across SSP scenarios, projected $PM_{2.5}$ concentrations generally decline relative to the historical baseline, with more pronounced reductions in the near- and mid-term and smaller net changes under stronger long-term warming. This trajectory is broadly consistent with climate-only assessments that report modest but non-negligible $PM_{2.5}$ responses to meteorological change. However, the spatial manifestation of these trends is highly uneven. While many areas—particularly in southwestern Taiwan—experience notable reductions, localized increases emerge in northeastern and southern coastal regions under specific GCM–SSP combinations.

The coexistence of overall $PM_{2.5}$ improvement with localized areas of potential increase is a critical finding. Regional mean concentrations alone obscure this spatial divergence, which arises from climate-driven changes in precipitation regimes, boundary-layer dynamics, and circulation patterns. Such heterogeneity implies that climate-related air-quality benefits are unlikely to be uniformly distributed and that some communities may face heightened exposure risks even as overall air quality improves. These results underscore the importance of high-resolution, spatially explicit assessments for identifying emerging pollution hotspots and informing geographically targeted mitigation and adaptation strategies.

Relative to most existing Geo-AI studies that focus on hindcasting historical $PM_{2.5}$, this work advances the field by incorporating down-scaled CMIP6 meteorology to enable future projection. Although the predictive performance of the selected model is comparable to that of many AOD-assisted retrospective models, the proposed framework avoids reliance on satellite aerosol products that suffer from substantial

data gaps in humid, cloud-prone island environments. In Taiwan, persistent cloud cover, high humidity, and frequent typhoon activity lead to pronounced missingness in AOD retrievals, limiting their utility for climate-conditioned applications. Within this context, the ability to generate stable, interpretable $PM_{2.5}$ projections without AOD inputs highlights the practical robustness of the Geo-AI approach adopted here.

4.2. Limitations and future studies

Despite these strengths, several sources of uncertainty warrant consideration. Although the selected CMIP6 global climate models (TaiESM1, MIROC6, and MPI-ESM1-2-LR) were statistically downscaled to improve representation of Taiwan's complex terrain and regional climate, previous evaluations indicate that GCMs may still exhibit biases in simulating extreme precipitation, particularly in monsoon-influenced regions. Because $PM_{2.5}$ responses in this framework are sensitive to precipitation and temperature anomalies, uncertainties in the frequency, intensity, and duration of extreme events may propagate into projected $PM_{2.5}$ patterns, especially in conditions dominated by wet scavenging or stagnation-driven accumulation.

TaiESM1 improves simulation of regional hydroclimate and diurnal rainfall cycles over Taiwan, yet uncertainties remain in representing typhoons, short-duration heavy rainfall, and coupled land–ocean–atmosphere processes. Observed increases in extreme precipitation indices across Taiwan may therefore not be fully captured by any single climate model. As a result, $PM_{2.5}$ anomalies associated with extreme meteorological events should be interpreted as indicative signals rather than deterministic predictions.

Another important limitation is that non-climatic predictors—including land use, population distribution, vegetation, and emission-related proxies—were held constant at baseline conditions. Comparing the study by (Grange and Carslaw, 2019), they isolate meteorological influences under the assumption that certain predictors remain relatively stable and used machine learning methods to predict air quality. Our Geo-AI framework does not assume strict baseline conditions. Instead, it integrates multi-source spatiotemporal predictors and leverages machine learning to capture nonlinear relationships between predictors and $PM_{2.5}$ concentrations. Moreover, SSP scenarios embedded assumptions about future socioeconomic development, only climate variables were incorporated into the $PM_{2.5}$ projections. Consequently, our results isolate the climate-driven component of future $PM_{2.5}$ change rather than representing comprehensive air-quality trajectories that also account for emission controls, energy transitions, or demographic shifts. This distinction is intentional: by holding non-climatic factors constant, the framework isolates the meteorological signal of climate change, providing a complementary perspective to emission-driven chemical transport model studies rather than a substitute for them.

Future research could integrate dynamic emission scenarios or hybridize Geo-AI projections with CTM constraints to further improve projection realism. Incorporating additional climate variables, such as boundary-layer height, solar radiation or wind-field changes, may also refine the characterization of climate– $PM_{2.5}$ interactions, particularly under extreme meteorological conditions.

4.3. Broader contributions

Beyond Taiwan, this study offers broader insights into $PM_{2.5}$ responses to climate change in humid subtropical, monsoon-influenced, and coastal–island environments. Regions such as southern China, Japan, the Korean Peninsula, Southeast Asia, the southeastern United States, eastern Australia, and parts of South America share key climatic and geographic characteristics relevant to this analysis, including strong seasonal precipitation variability, complex land–sea interactions, and pronounced topographic heterogeneity. In these settings, climate-driven shifts in temperature and precipitation can exert nonlinear—and

sometimes counterintuitive—effects on particulate pollution that may not track emission trends in a straightforward manner.

The observed pattern of overall PM_{2.5} reduction accompanied by localized hotspots likely reflects a broader climate–air quality response in similarly structured regions. This finding highlights the limitations of relying solely on regional mean PM_{2.5} trends and reinforces the need for fine-scale, climate-conditioned assessments that can identify spatially heterogeneous risks and support targeted adaptation planning.

Methodologically, this work demonstrates that Geo-AI models can be effectively coupled with statistically downscaled CMIP6 projections to generate high-resolution, climate-conditioned PM_{2.5} estimates without requiring detailed future emission inventories or computationally intensive chemical transport simulations. The proposed framework thus provides a practical bridge between physically based modeling and purely statistical extrapolation, enabling scalable assessment of climate–pollution interactions in data-rich regions. Taken together, this study illustrates that Geo-AI, when integrated with climate projections, can function not only as a high-resolution mapping tool but as a forward-looking analytical framework for anticipating climate-driven air-quality risks in complex, climate-sensitive environments.

5. Conclusions

This study develops a high-resolution Geo-AI framework linking ground observations, geospatial predictors, and downscaled climate projections to assess climate-driven PM_{2.5} dynamics. The results suggest that temperature and precipitation significantly influence PM_{2.5} patterns, leading to potential overall reductions but increasing spatial heterogeneity and localized hotspots under future warming.

These findings provide actionable insights for climate-resilient air quality management and urban planning strategies, particularly in regions where climate benefits are uneven. However, the projections should be interpreted as climate-driven sensitivity under fixed emission conditions, as future emission trends, policy interventions, and socio-economic changes remain uncertain and may substantially alter outcomes.

The proposed framework provides a computationally efficient and transferable approach for climate-informed air-quality assessment, particularly in monsoon-influenced and topographically complex regions. Future integration of dynamic emission scenarios and process-based constraints would further improve robustness and policy relevance for long-term environmental planning.

CRedit authorship contribution statement

Pei-Yi Wong: Data curation, Formal analysis, Investigation, Methodology, Project administration, Software, Validation, Writing – original draft, Writing – review & editing. **Wei-Lun Hsu:** Data curation, Investigation, Methodology, Software, Writing – review & editing. **Chieh-Ying Chen:** Visualization, Writing – review & editing. **Chien-Fei Chen:** Conceptualization, Resources, Writing – review & editing. **Chia-Wei Hsu:** Writing – review & editing. **Shih-Chun Candice Lung:** Conceptualization, Resources, Writing – review & editing. **Ta-Chih Hsiao:** Conceptualization, Resources, Writing – review & editing. **Chia-Pin Yu:** Conceptualization, Resources, Writing – review & editing. **Wan-Yu Liu:** Conceptualization, Resources, Writing – review & editing. **Pau-Chung Chen:** Conceptualization, Resources, Writing – review & editing. **Wei Jie Seow:** Conceptualization, Resources, Writing – review & editing. **Chien-Hao Sung:** Writing – review & editing. **Chih-Da Wu:** Conceptualization, Funding acquisition, Resources, Supervision, Visualization, Writing – review & editing.

Declaration of competing interest

■ The authors declare the following financial interests/personal relationships which may be considered as potential competing

interests: Chih-Da Wu reports financial support was provided by National Science and Technology Council, Taiwan. Chih-Da Wu reports financial support was provided by Ministry of Education (MOE), Taiwan. Chih-Da Wu reports financial support was provided by Kaohsiung Medical University, Taiwan. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This study was supported by the National Science and Technology Council, Taiwan (NSTC 115-2119-M-006-006 - ; 115-2321-B-182-002 - ; 115-2420-H-006-002 -HS2 ; 114-2123-M-001-006-; 114-2121-M-006-001; 114-2811-M-001-066), and the Higher Education Sprout Project by the Ministry of Education (MOE) in Taiwan. This work was supported partially by the Research Center for Precision Environmental Medicine, Kaohsiung Medical University, Kaohsiung, Taiwan from The Featured Areas Research Center Program within the framework of the Higher Education Sprout Project by the Ministry of Education (MOE) in Taiwan and by Kaohsiung Medical University Research Center Grant (KMU-TC114A201). This research was also funded by the Chang Gung University of Science and Technology Foundation (ZRRPF6P0011, ZRRPF6Q0011), and Chang Gung Medical Research Program Foundation (CORPF6P0031).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jenvman.2026.130232>.

Data availability

Data will be made available on request.

References

- Amnuaylojaroen, T., Surapipith, V., Macatangay, R.C., 2022. Projection of the near-future PM_{2.5} in Northern peninsular Southeast Asia under RCP8.5. *Atmosphere* 13 (2), 305. <https://doi.org/10.3390/atmos13020305>.
- Babaan, J., Hsu, F.-T., Wong, P.-Y., Chen, P.-C., Guo, Y.-L., Lung, S.-C.C., Chen, Y.-C., Wu, C.-D., 2023. A Geo-AI-based ensemble mixed spatial prediction model with fine spatial-temporal resolution for estimating daytime/nighttime/daily average ozone concentrations variations in Taiwan. *J. Hazard Mater.* 446, 130749. <https://doi.org/10.1016/j.jhazmat.2023.130749>.
- Chen, H.-W., Chen, C.-Y., Chang, T.-W., Lin, G.-Y., 2023. Spatial-temporal evaluation of PM_{2.5} concentration for health risk reduction strategy development in a basin with different weather patterns. *Atmos. Pollut. Res.* 14 (10), 101884. <https://doi.org/10.1016/j.apr.2023.101884>.
- Chen, H.-C., Jin, F.-F., 2021. Simulations of ENSO phase-locking in CMIP5 and CMIP6. *J. Clim.* 34 (12), 5135–5149. <https://doi.org/10.1175/JCLI-D-20-0874.1>.
- Chuang, M.-T., Chou, C.C.K., Lin, N.-H., Takami, A., Hsiao, T.-C., Lin, T.-H., Fu, J.S., Pani, S.K., Lu, Y.-R., Yang, T.-Y., 2017. A simulation study on PM_{2.5} sources and meteorological characteristics at the Northern tip of Taiwan in the early stage of the Asian haze period. *Aerosol Air Qual. Res.* 17 (12), 3166–3178. <https://doi.org/10.4209/aaqr.2017.05.0185>.
- Demuzere, M., Kittner, J., Martilli, A., Mills, G., Moede, C., Stewart, Iain D., van Vliet, J., Bechtel, B., 2022. A global map of local climate zones to support earth system modelling and urban-scale environmental science. *Earth Syst. Sci. Data* 14, 3835–3873. <https://doi.org/10.5194/essd-14-3835-2022>.
- Donkelaar, A., Martin, R., Brauer, M., Boys, B., 2014. Use of satellite observations for long-term exposure assessment of global concentrations of fine particulate matter. *Environ. Health Perspect.* 123, 135–143. <https://doi.org/10.1289/ehp.1408646>.
- Döscher, R., Acosta, M., Alessandri, A., Anthoni, P., Arneth, A., Arsouze, T., Bergman, T., Bernardello, R., Boussetta, S., Caron, L.P., Carver, G., Castrillo, M., Catalano, F., Cvijanovic, I., Davini, P., Dekker, E., Doblas, F., Docquier, D., Echevarria, P., Zhang, Q., 2021. The EC-Earth3 Earth system model for the climate model intercomparison project 6. *Geosci. Model Dev.* 2973–3020. <https://doi.org/10.5194/gmd-2020-446>, 15(7).
- Feng, H., Zou, B., Wang, J., Gu, X., 2019. Dominant variables of global air pollution-climate interaction: geographic insight. *Ecol. Indic.* 99, 251–260. <https://doi.org/10.1016/j.ecolind.2018.12.038>.
- Grange, Stuart K., Carslaw, David C., 2019. Using meteorological normalisation to detect interventions in air quality time series. *Science of The Total Environment* 653, 578–588. <https://doi.org/10.1016/j.scitotenv.2018.10.344>. <https://linkinghub.elsevier.com/retrieve/pii/S004896971834244X>.

- Hsu, C.-Y., Lin, T.-W., Babaan, J.B., Asri, A.K., Wong, P.-Y., Chi, K.-H., Ngo, T.H., Yang, Y.-H., Pan, W.-C., Wu, C.-D., 2023. Estimating the daily average concentration variations of PCDD/Fs in Taiwan using a novel Geo-AI based ensemble mixed spatial model. *J. Hazard Mater.* 458, 131859. <https://doi.org/10.1016/j.jhazmat.2023.131859>.
- Kumar, R.P., Jahan, A., Singh, R., Kumar, P., Bag, R., Bhatla, R., Ambade, B., Dumka, U. C., 2025a. Spatio-temporal analysis of air pollution and meteorological influences in Western Uttar Pradesh using geospatial techniques: insights for policy and management. *Int. J. Rem. Sens.* 1–28. <https://doi.org/10.1080/01431161.2025.2529601>.
- Kumar, R.P., Prakash, A., Singh, R., Kumar, P., 2024b. Machine learning-based prediction of hazards fine PM_{2.5} concentrations: a case study of Delhi, India. *Discover Geoscience* 2 (1), 34. <https://doi.org/10.1007/s44288-024-00043-z>.
- Kumar, R.P., Rana, R., Choudhary, A., Singh, R., 2025b. Spatiotemporal variability and source attribution of PM_{2.5}/PM₁₀ ratios: aerosol type classification and AQI evaluation across seventy monitoring stations in Delhi and Haryana, India. *Phys. Chem. Earth, Parts A/B/C* 140, 104005. <https://doi.org/10.1016/j.pce.2025.104005>.
- Kumar, R.P., Singh, R., Kumar, P., Kumar, R., Nahid, S., Singh, S.K., Nijjar, C.S., 2024a. Aerosol-PM_{2.5} dynamics: in-situ and satellite observations under the influence of regional crop residue burning in post-monsoon over Delhi-NCR, India. *Environ. Res.* 255, 119141. <https://doi.org/10.1016/j.envres.2024.119141>.
- Kumar, R.P., Choudhary, A., 2025. Spatiotemporal dynamics of air pollutants and human health burden in Bihar, India: a multi-source assessment for effective air quality management. *Phys. Chem. Earth, Parts A/B/C* 140, 104047. <https://doi.org/10.1016/j.pce.2025.104047>.
- Lai, H.-C., Lin, M.-C., 2020. Characteristics of the upstream flow patterns during PM_{2.5} pollution events over a complex island topography. *Atmos. Environ.* 227, 117418. <https://doi.org/10.1016/j.atmosenv.2020.117418>.
- Lee, Y.-Y., Hsieh, Y.-K., Chang-Chien, G.-P., Wang, W., 2019. Characterization of the air quality index in Southwestern Taiwan. *Aerosol Air Qual. Res.* 19 (4), 749–785. <https://doi.org/10.4209/aaqr.2019.02.0080>.
- Li, L., Zheng, M., Zhang, J., Li, C., Ren, Y., Jin, X., Chen, J., 2023. Effects of green infrastructure on the dispersion of PM_{2.5} and human exposure on urban roads. *Environ. Res.* 223, 115493. <https://doi.org/10.1016/j.envres.2023.115493>.
- Lin, Lee-Yaw, Lin, Cheng-Ting, Chen, Yung-Ming, Cheng, Chao-Tzuen, Li, Hsin-Chi, et al., 2022. The Taiwan Climate Change Projection Information and Adaptation Knowledge Platform: A Decade of Climate Research. *Water* 14 (3), 358. <https://doi.org/10.3390/w14030358>. <https://www.mdpi.com/2073-4441/14/3/358>. (Accessed 26 January 2022).
- Liu, H., Yan, G., Duan, Z., Chen, C., 2021. Intelligent modeling strategies for forecasting air quality time series: a review. *Appl. Soft Comput.* 102, 106957. <https://doi.org/10.1016/j.asoc.2020.106957>.
- Lovato, T., Peano, D., Butenschön, M., Matera, S., Iovino, D., Scoccimarro, E., Fogli, P. G., Cherchi, A., Bellucci, A., Gualdi, S., Masina, S., Navarra, A., 2022. CMIP6 simulations with the CMCC Earth system model (CMCC-ESM2). *J. Adv. Model. Earth Syst.* 14. <https://doi.org/10.1029/2021MS002814>.
- Lundberg, S.M., Lee, S.I., 2017. A unified approach to interpreting model predictions. *Adv. Neural Inf. Process. Syst.* 30, 4765–4774. <https://doi.org/10.48550/arXiv.1705.07874>.
- Luo, D., Sommar, J.O., Liu, W., Zhao, H., 2025. Wavelet multiscale analysis reveals PM_{2.5} and PM₁₀ temporal dynamics and meteorological coupling in Guiyang, a representative Karst Highland city. *ACS Omega* 10 (47), 57641–57656. <https://doi.org/10.1021/acsomega.5c09015>.
- Lyu, B., Huang, R., Wang, X., Wang, W., Hu, Y., 2025. FastCTM (v1.0): atmospheric chemical transport modelling with a principle-informed neural network for air quality simulations. *Geosci. Model Dev.* 18 (18), 6295–6312. <https://doi.org/10.5194/gmd-18-6295-2025>.
- Ma, Y., Xin, J., Wang, Z., Tian, Y., Wu, L., Tang, G., Zhang, W., de Arellano, J.V.-G., Zhao, D., Jia, D., Ren, Y., Gao, Z., Shen, P., Ye, J., Martin, S.T., 2022. How do aerosols above the residual layer affect the planetary boundary layer height? *Sci. Total Environ.* 814, 151953. <https://doi.org/10.1016/j.scitotenv.2021.151953>.
- Megaritis, A.G., Fountoukis, C., Charalampidis, P.E., Denier van der Gon, H.A.C., Pilinis, C., Pandis, S.N., 2014. Linking climate and air quality over Europe: effects of meteorology on PM_{2.5} concentrations. *Atmos. Chem. Phys.* 14 (18), 10283–10298. <https://doi.org/10.5194/acp-14-10283-2014>.
- O'Neill, B.C., Tebaldi, C., van Vuuren, D.P., Eyring, V., Friedlingstein, P., Hurtt, G., Knutti, R., Krieger, E., Lamarque, J.F., Lowe, J., Meehl, G.A., Moss, R., Riahi, K., Sanderson, B.M., 2016. The scenario model intercomparison project (ScenarioMIP) for CMIP6. *Geosci. Model Dev.* 9 (9), 3461–3482. <https://doi.org/10.5194/gmd-9-3461-2016>.
- Peng, L.-C., Lin, Y.-P., Chen, G.-W., Lien, W.-Y., 2019. Climate change impact on spatiotemporal hotspots of hydrologic ecosystem services: a case study of chinan catchment, Taiwan. *Water* 11 (4), 867. <https://www.mdpi.com/2073-4441/11/4/867>.
- Srbínovska, M., Andova, V., Mateska, A.K., Krstevska, M.C., 2021. The effect of small green walls on reduction of particulate matter concentration in open areas. *J. Clean. Prod.* 279, 123306. <https://doi.org/10.1016/j.jclepro.2020.123306>.
- Taiwan Climate Change Projection Information and Adaptation Knowledge Platform (TCCIP), 2024. Taiwan AR6 climate projections database. National Science and Technology Center for Disaster Reduction. <https://tccip.ncdr.nat.gov.tw>.
- Wang, J., Chen, Y., Liao, W., He, G., Tett, S.F.B., Yan, Z., Zhai, P., Feng, J., Ma, W., Huang, C., Hu, Y., 2021a. Anthropogenic emissions and urbanization increase risk of compound hot extremes in cities. *Nat. Clim. Change* 11 (12), 1084–1089. <https://doi.org/10.1038/s41558-021-01196-2>.
- Wang, Y.-C., Hsu, H.-H., Chen, C.-A., Tseng, W.-L., Hsu, P.C., Lin, C.W., Chen, Y.-L., Jiang, L.-C., Lee, Y.C., Liang, H.-C., Chang, W.M., Lee, W.-L., Shiu, C.J., 2021b. Performance of the Taiwan Earth system model in simulating climate variability compared with observations and CMIP6 model simulations. Performance of the Taiwan Earth system model in simulating climate variability compared with observations and CMIP6 model simulations. *J. Adv. Model. Earth Syst.* 13 (7). <https://doi.org/10.1029/2020MS002353>.
- West, J.J., Smith, S.J., Silva, R.A., Naik, V., Zhang, Y., Adelman, Z., Fry, M.M., Anenberg, S., Horowitz, L.W., Lamarque, J.-F., 2013. Co-benefits of mitigating global greenhouse gas emissions for future air quality and human health. *Nat. Clim. Change* 3 (10), 885–889. <https://doi.org/10.1038/nclimate2009>.
- Westervelt, D.M., Horowitz, L.W., Naik, V., Tai, A.P.K., Fiore, A.M., Mauzerall, D.L., 2016. Quantifying PM_{2.5}-meteorology sensitivities in a global climate model. *Atmos. Environ.* 142, 43–56. <https://doi.org/10.1016/j.atmosenv.2016.07.040>.
- Wong, P.-Y., Lee, H.-Y., Chen, L.-J., Chen, Y.-C., Chen, N.-T., Lung, S.-C.C., Su, H.-J., Wu, C.-D., Laurent, J.G.C., Adamkiewicz, G., Spengler, J.D., 2022. An alternative approach for estimating large-area indoor PM_{2.5} concentration – a case study of schools. *Build. Environ.* 219, 109249. <https://doi.org/10.1016/j.buildenv.2022.109249>.
- Wong, P.-Y., Lee, H.-Y., Chen, Y.-C., Zeng, Y.-T., Chern, Y.-R., Chen, N.-T., Candice Lung, S.-C., Su, H.-J., Wu, C.-D., 2021. Using a land use regression model with machine learning to estimate ground level PM_{2.5}. *Environ. Pollut.* 277, 116846. <https://doi.org/10.1016/j.envpol.2021.116846>.
- Wong, P.-Y., Su, H.-J., Candice Lung, S.-C., Liu, W.-Y., Tseng, H.-T., Adamkiewicz, G., Wu, C.-D., 2024a. Explainable geospatial-artificial intelligence models for the estimation of PM_{2.5} concentration variation during commuting rush hours in Taiwan. *Environ. Pollut.* 349, 123974. <https://doi.org/10.1016/j.envpol.2024.123974>.
- Wong, P.-Y., Su, H.-J., Chao, H.J., Pan, W.-C., Tsai, H.-J., Yao, T.-C., Liu, W.-Y., Lung, S.-C.C., Adamkiewicz, G., Wu, C.-D., 2024b. An innovative Geo-AI approach in estimating high-resolution urban ambient fungal spore variations. *Earth Syst. Environ.* <https://doi.org/10.1007/s41748-024-00535-5>.
- Wong, P.-Y., Su, H.-J., Lung, S.-C.C., Wu, C.-D., 2023. An ensemble mixed spatial model in estimating long-term and diurnal variations of PM_{2.5} in Taiwan. *Sci. Total Environ.* 866, 161336. <https://doi.org/10.1016/j.scitotenv.2022.161336>.
- Xiao, F., Yang, M., Fan, H., Fan, G., Al-qaness, M.A.A., 2020. An improved deep learning model for predicting daily PM_{2.5} concentration. *Sci. Rep.* 10 (1), 20988. <https://doi.org/10.1038/s41598-020-77757-w>.
- Yang, Y., Liao, H., Lou, S., 2016. Increase in winter haze over eastern China in recent decades: roles of variations in meteorological parameters and anthropogenic emissions. *J. Geophys. Res. Atmos.* 121 (21), 13050–13065. <https://doi.org/10.1002/2016JD025136>.
- Zaman, N.A.F.K., Kanniah, K.D., Kaskaoutis, D.G., Latif, M.T., 2021. Evaluation of machine learning models for estimating PM_{2.5} concentrations across Malaysia. *Appl. Sci.* 11 (16), 7326. <https://doi.org/10.3390/app11167326>.